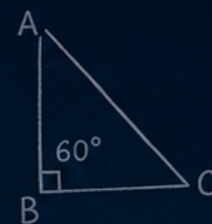


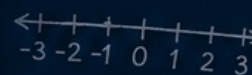
$$y = 2x + 3$$

$$\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

$$2(x - 3) = 14$$



$$a^2 + b^2 = c^2$$



★ VOLUME 1 ★

EMERGENCY SUB PLANS

for MIDDLE SCHOOL MATH

GRADES 6-8

KEEP LEARNING ON TRACK WHEN PLANS CHANGE



NO-PREP ACTIVITIES



READY-TO-USE RESOURCES



PERFECT FOR EMERGENCY LESSONS



GREAT FOR SUBSTITUTE TEACHERS



PRINT OR DIGITAL USE

RATIO & PROPORTIONS

Solve each problem.

- The ratio of boys to girls in a class is 3:5. If there are 24 girls, how many boys are in the class?



- Simplify the ratio 18:24.

SOLVE FOR x

Solve each equation.

$$1. 2x + 5 = 17$$

$$2. 3(x - 4) = 21$$

$$3. \frac{x}{4} + 3 = 8$$

$$4. 5x - 7 = 28$$

PLOT & ANALYZE

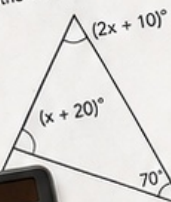
The graph shows the number of skateboards sold each week.



- How many skateboards were sold in Week 4?
- Between which weeks did sales decrease?

GEOMETRY

Find the value of x.



x =

MATH EMERGENCY KIT

ORDER OF OPERATIONS

Evaluate each expression.

- $12 + 6 \times 2 - 4$
- $(16 - 8) \div 3 + 7$
- $2^4 + 3 \times (8 - 5)$
- $24 \div (2 + 2) + 5$

YOUR CLASSROOM LIFESAVER



MADE BY
TEACHERS
for
TEACHERS

Set this up while you are well

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a morning when you are not.

The one job you have today

Print pages 3–6 and leave them in a clearly labeled folder where a substitute will find them — top drawer, on the desk, or at the front office. Everything a sub needs is in those four pages.

You do not need to print the lessons ahead of time. The sub picks one on the day and copies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a folder or binder. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. Those are the pages a substitute reads first.
- 3 **Tell your department chair where it lives.** On a real sick day you will not be answering the phone, so somebody else needs to know.
- 4 **Cross off lessons as you use them.** There is a tracker on page 4. Two years of absences will never repeat if you keep it current.
- 5 **Optional: leave a class roster** and a note about students who need a reader, extra time, or specific seating.

Why nothing here needs technology

On a sub day the Chromebook cart is often locked, the login server is down, or nobody knows the room code. Every lesson runs on paper and a pencil.

Calculators are optional throughout, and each lesson says where one actually helps.

Why it is not busy work

Every lesson targets a real Grade 6–8 standard and produces graded evidence you can put in a folder.

Away for a week? Running four of these in sequence is a defensible unit, not a holding pattern.

About the answer keys

Every key sits on the **last page of its lesson** and shows the **working**, not just the answer. That is what lets a sub who has not taught math in years respond when a student asks “*but why?*”

You do not need to be a math teacher

You have walked into a middle school math class. You may not have done algebra since high school. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to be able to do the math. Every lesson is a paper worksheet with a complete answer key that shows the working. Your job is to hand it out, keep the room settled, and check answers against the key. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for Grades 6, 7 or 8. If you cannot decide, use Lesson 10.
- 2 **Copy the student pages only.** Each lesson tells you exactly which page numbers. One per student.
- 3 **Keep the teacher page and the answer key.** Do not copy those.
- 4 **Read the teacher page once** before the bell. Two minutes, and it includes the exact words you can use to start.

If a student says “we already did this”

Ask them to prove it by finishing the challenge section on the last page. If they genuinely have, switch them to a different lesson — there are ten, and they will not have done all of them.

If they finish early

Every lesson has a challenge task on its final student page. There is also a discussion question on the teacher page you can run with the whole class for the last five minutes.

What to leave behind

Collect the sheets. Fill in the handover note on the last page of this pack — ninety seconds, and it tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Choose your lesson

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	The Answer Is Wrong	Order of operations	★	50	7–9
02	Fractions, Decimals & Percents	Converting & comparing	★★	55	12–14
03	Best Buy	Ratios & unit rates	★★	55	17–19
04	Above and Below Zero	Integer operations	★★	50	22–24
05	Keep It Balanced	Solving equations	★★★	55	27–29
06	Design a Room	Area, perimeter & cost	★★	60	32–34
07	How Graphs Lie	Mean, median, mode & data	★	55	37–39
08	Is the Game Fair?	Probability	★★	55	42–44
09	The Coordinate Plane	Plotting & transformations	★★	55	47–49
10	Tips, Discounts & Sales Tax	Percent in real life	★	60	52–54

Easiest to supervise

01, 07 and 10. Real-world and error-spotting tasks. Students have opinions, so the room stays busy.

Best for a quiet class

02, 04 and 05. Straight practice with definite answers. Fast finishers will race — that is what the challenges are for.

Best for a restless class

03, 06 and 08. Shopping, floor plans and games of chance. Noisier, but productively so.

Used-lesson tracker • for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Order of operations			
02 — Fractions & percents			
03 — Ratios & unit rates			
04 — Integers			
05 — Equations			
06 — Area & perimeter			
07 — Data & graphs			
08 — Probability			
09 — Coordinate plane			
10 — Percent			

The first ten minutes

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in and take a seat. Bags off the desks. You will not need a Chromebook today – everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is out today. My name is _____. This is a real math lesson, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their own teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the first panel together	Have a student read the worked example aloud
10–40	Students work through the tasks	Circulate. Check against the answer key
40–50	Challenge task for early finishers	Point them to the last student page
50–55	Collect sheets, pack up	Fill in the handover note

"I don't get it"

Almost always means *I don't know how to start*. Point at the worked example on the first page and ask them to copy its first line with their own numbers.

That single move unsticks most students without you needing to explain anything.

"Can I use a calculator?"

Unless the teacher page says otherwise, yes for the arithmetic and no for the reasoning tasks. Each teacher page states which.

If in doubt, allow it. A student doing the thinking with a calculator is better than one doing nothing.

Copying reminder

Student pages only. Each teacher page gives the exact range. The **answer key is always the last page** of the lesson — keep it.

The Answer Is Wrong

2 MINUTES

Students evaluate expressions using order of operations, then switch roles and hunt for the mistake in somebody else's work. Finding errors is far more engaging than doing sums, and it produces the same practice.

You do not need to know this subject

Every answer is a single number and every one is in the key with the working shown. If you can compare two numbers, you can grade this lesson.

Photocopy these pages only

Pages 7–9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are not needed but do no harm.

WHAT TO SAY TO START

“Every one of these problems has one correct answer, and most of you will get some of them wrong for exactly the same reason. Today you are going to find out what that reason is, and then go hunting for it in somebody else's work.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read PEMDAS and the worked example	Do the worked example on the board with them
12–24	Task 1.1 — evaluate expressions	Circulate. Every answer is in the key
24–38	Task 1.2 — find the error	The best part. Let them argue
38–46	Task 1.3 — add the parentheses	Harder than it looks
46–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is it PEMDAS or BODMAS?”

Same rule, different name. Multiply and divide are done together left to right, and so are add and subtract.

“Why is $6 \div 2 \times 3$ equal to 9 and not 1?”

Divide and multiply have equal rank, so you work left to right: $6 \div 2 = 3$, then $3 \times 3 = 9$. This is the most common argument in this lesson.

“Do I need a calculator?”

No. All the arithmetic is small on purpose. Allow one if a student is stuck on the arithmetic rather than the order.

If they finish early

Task 1.4 asks students to invent an expression that traps a classmate. Beyond that, ask them to write one expression that gives a different answer depending on whether you go left to right or right to left, and explain which is correct.

Curriculum focus • for the regular teacher's records

Grade 6–7: numeric expressions and order of operations. Discussion question for the last five minutes: *“If everyone did the operations in whatever order they liked, what would happen to math?”*

One expression, one answer

14 MIN

NAME _____

CLASS _____

DATE _____

The order is not a suggestion

Everyone on Earth agrees to do operations in the same order. Without that agreement, the same expression would mean different things to different people.

The two rules people forget: multiply and divide have **equal** rank, and so do add and subtract. When two operations have equal rank, work **left to right**.

P Parentheses ()

E Exponents 3^2

MD Multiply & Divide ← left to right

AS Add & Subtract ← left to right

Worked example • $4 + 3 \times (8 - 6)^2 \div 2$

$$\begin{aligned}
 &4 + 3 \times (8 - 6)^2 \div 2 \\
 &= 4 + 3 \times 2^2 \div 2 \quad \leftarrow \text{parentheses first} \\
 &= 4 + 3 \times 4 \div 2 \quad \leftarrow \text{exponent} \\
 &= 4 + 12 \div 2 \quad \leftarrow \text{multiply (leftmost)} \\
 &= 4 + 6 \\
 &= 10
 \end{aligned}$$

Notice line 4. Multiply came before divide only because it was further **left**, not because M comes before D.

Task 1.1 • Evaluate each one

DO

Show at least two lines of working for each.

#	EXPRESSION	WORKING	ANSWER
a	$12 - 3 \times 2$		
b	$(12 - 3) \times 2$		
c	$20 \div 4 \times 5$		
d	$6 + 2^3 - 4$		
e	$3 \times (4 + 2)^2$		
f	$18 - (6 \div 2) \times 4$		

Somebody else's mistake

16 MIN

Task 1.2 • Find the error

DETECTIVE

Each solution below is wrong. Circle the **first** line where it goes wrong, say what the student did, and give the correct answer.

Alex • $8 + 4 \times 3$
 $= 12 \times 3$
 $= 36$

What went wrong?

Correct answer: _____

Bri • $20 - 6 \div 2$
 $= 20 - 3$
 $= 17$

What went wrong?

Correct answer: _____

Chris • $24 \div 6 \times 2$
 $= 24 \div 12$
 $= 2$

What went wrong?

Correct answer: _____

Dev • $2 + 3^2$
 $= 5^2$
 $= 25$

What went wrong?

Correct answer: _____

Task 1.2b • Harder expressions

DO

These have exponents and nested parentheses. Work from the inside out.

#	EXPRESSION	WORKING	ANSWER
g	$2 \times (3 + 4^2)$		
h	$(10 - 2)^2 \div 4$		
i	$5 + 2 \times (9 - 3) \div 3$		
j	$((6 + 2) \times 3) - 4^2$		

Make it true

16 MIN

Task 1.3 • Add the parentheses

CHALLENGE

Each equation below is **false** as written. Add one pair of parentheses to make it true.

#	MAKE THIS TRUE	YOUR VERSION
a	$4 + 6 \times 2 = 20$	
b	$18 - 6 \div 3 = 4$	
c	$5 + 3 \times 4 - 2 = 30$	
d	$24 \div 4 + 2 = 4$	

Task 1.4 • Build a trap

CREATE

Write an expression using at least **four** numbers and **three** different operations, designed so that somebody who ignores the order gets it wrong.

My expression:

The correct answer is _____

Somebody who works strictly left to right would get _____

Swap with a partner and solve theirs. Did you fall for it? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. If everyone did the operations in whatever order they liked, what would happen to math?

Answer key

The Answer Is Wrong

ANSWERS

Task 1.1 • Answers with working

- a. $12 - 6 = 6$
- b. $9 \times 2 = 18$
- c. $5 \times 5 = 25$
- d. $6 + 8 - 4 = 10$
- e. $3 \times 36 = 108$
- f. $18 - 12 = 6$

c is the argument starter. $20 \div 4 \times 5$ goes left to right, so $5 \times 5 = 25$. Students who do 4×5 first get 1.

Task 1.3 • Where the parentheses go

- a. $(4 + 6) \times 2 = 20$
- b. $(18 - 6) \div 3 = 4$
- c. $(5 + 3) \times 4 - 2 = 30$
- d. $24 \div (4 + 2) = 4$

Accept any placement that produces the stated result — for c, $5 + 3 \times (4 - 2)$ does not work, so check the arithmetic rather than the shape.

Task 1.2b • Harder expressions

- g. $2 \times (3 + 16) = 2 \times 19 = 38$
- h. $8^2 \div 4 = 64 \div 4 = 16$
- i. $5 + 2 \times 6 \div 3 = 5 + 12 \div 3 = 5 + 4 = 9$
- j. $(8 \times 3) - 16 = 24 - 16 = 8$

Task 1.2 • The four errors

STUDENT	THE ERROR	CORRECT
Alex	Added before multiplying. Multiplication outranks addition, so $4 \times 3 = 12$ comes first.	20
Bri	This one is actually correct in method — but the final subtraction is wrong. $20 - 3 = 17$ is right. See note below.	17
Chris	Multiplied before dividing. They have equal rank, so work left to right: $24 \div 6 = 4$, then $4 \times 2 = 8$.	8
Dev	Added before applying the exponent. $3^2 = 9$ comes first, then $2 + 9$.	11

Read this before grading Task 1.2

Bri's solution is correct. That is deliberate. Students who mark all four as wrong have assumed the question rather than checked it — which is exactly the habit this lesson is trying to break. Award full credit to any student who says Bri is right, and use it as the discussion point if the class is quick.

Task 1.4 • Grading the traps

Full credit for any expression with four or more numbers and three different operations where the correct answer genuinely differs from the strict left-to-right answer. A good example is $2 + 3 \times 4 - 6 \div 2$: correct answer **11**, left-to-right answer **4**.

Fractions, Decimals & Percents

2 MINUTES

Students convert between the three forms, then use them to settle real comparisons — which discount is bigger, which test score is better. The conversions become useful rather than abstract.

You do not need to know this subject

Every answer is a definite number and the key shows the conversion for each one. The only skill you need is comparing your student's answer to the printed one.

Photocopy these pages only

Pages 12–14

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. A calculator is helpful for Task 2.1 and fine to allow.

WHAT TO SAY TO START

“One half. Nought point five. Fifty percent. Three different ways of writing exactly the same amount. Today you will learn to move between them quickly, and then use that to win an argument about which store has the better deal.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the conversion triangle together	Walk through the worked example on the board
14–28	Task 2.1 — the conversion table	Every cell is in the key
28–40	Task 2.2 — ordering mixed forms	They must convert to compare. That is the point
40–50	Task 2.3 — which is the better deal	Real-world reasoning
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Do I move the decimal left or right?”

Percent to decimal, move left two places. Decimal to percent, move right two. The reminder box on the first student page says exactly this.

“Can I use a calculator?”

Yes for the division in Task 2.1. The thinking is in knowing what to divide, not in doing it.

“Is 0.4 bigger than 0.25?”

Yes, and this trips up more students than anything else in the lesson. 0.4 is 0.40. Suggest they add the trailing zero so both have the same number of places.

If they finish early

Task 2.4 asks students to find the percent increase on a real price. Beyond that, ask them which is larger: $\frac{1}{3}$ or 0.333, and to justify it. The answer surprises them.

Curriculum focus • for the regular teacher's records

Grade 6–7: equivalent fractions, decimals and percents; comparing and ordering rational numbers. Discussion question: “Stores advertise ‘% off’ and ‘35% off’. Which sounds bigger, and which is bigger?”

Three ways to say the same thing

16 MIN

NAME _____

CLASS _____

DATE _____

Moving between the three forms

FRACTION → DECIMAL

Divide top by bottom.
 $\frac{3}{4} = 3 \div 4 = 0.75$

DECIMAL → PERCENT

Move the point **right** two.
 $0.75 \rightarrow 75\%$

PERCENT → FRACTION

Put it over 100, simplify.
 $75\% = \frac{75}{100} = \frac{3}{4}$

Worked example • write $\frac{3}{8}$ all three ways

Fraction: $\frac{3}{8}$

Decimal: $3 \div 8 = 0.375$

Percent: $0.375 \rightarrow$ move right two $\rightarrow 37.5\%$

Check it makes sense: $\frac{3}{8}$ is a bit less than half, and 37.5% is a bit less than 50% . **Always sanity-check against a half.**

Task 2.1 • Complete the table

DO

FRACTION	DECIMAL	PERCENT
$\frac{1}{2}$		
	0.25	
		10%
$\frac{3}{5}$		
	0.125	
		80%
$\frac{7}{10}$		
	0.05	

Now compare them

18 MIN

Task 2.2 • Smallest to largest

ORDER

You cannot compare these until they are all in the same form. Convert them all to **decimals** first, then order them.

SET	THE VALUES				AS DECIMALS	SMALLEST → LARGEST
A	0.4	1/2	45%	0.35		
B	2/5	0.25	30%	1/3		
C	0.7	3/4	68%	7/10		

Careful: is 0.4 bigger or smaller than 0.35? Write both with the same number of decimal places before you decide.

Task 2.3 • Which is the better deal?

DECIDE

Same jacket, three stores. Show your working, then circle the cheapest.

STORE	ORIGINAL	OFFER	WORKING	FINAL PRICE
Northgate	\$60	1/2 off		
Riverside	\$60	30% off		
Eastway	\$60	0.35 off		

Which offer **sounds** biggest, and which one actually **is** biggest? _____

Task 2.3b • Quick percents

DRILL

Find each amount. Use the shortcut: 10% is the decimal point moved one place left.

FIND	ANSWER	FIND	ANSWER
10% of \$60		20% of \$45	
25% of \$80		5% of \$120	
50% of \$37		75% of \$32	

Percents that matter

16 MIN

Task 2.4 • Which score is better?

COMPARE

Four students, four different tests. Convert each to a percent to find who did best.

STUDENT	SCORE	WORKING	PERCENT	RANK
Maya	17 out of 20			
Jordan	21 out of 25			
Priya	33 out of 40			
Sam	13 out of 16			

Task 2.5 • The trap question

CHALLENGE

a. Which is larger, $\frac{1}{3}$ or 0.333? Show why.

b. A jacket costs \$80. The price goes up 10%, then goes back down 10%. Is it back to \$80? Show your working.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. A store advertises “ $\frac{1}{3}$ off” and another advertises “35% off”. Which sounds bigger, and which is bigger?

Answer key

Fractions, Decimals & Percents

ANSWERS

Task 2.1 • The completed table

FRACTION	DECIMAL	PERCENT
1/2	0.5	50%
1/4	0.25	25%
1/10	0.1	10%
3/5	0.6	60%
1/8	0.125	12.5%
4/5	0.8	80%
7/10	0.7	70%
1/20	0.05	5%

Task 2.2 • Ordering

A: 0.35, 0.4, 45%, 1/2

(0.35 • 0.4 • 0.45 • 0.5)

B: 0.25, 30%, 1/3, 2/5

(0.25 • 0.3 • 0.333 • 0.4)

C: 68%, 0.7, 7/10, 3/4

(0.68 • 0.7 • 0.7 • 0.75)

In set C, **0.7 and 7/10 are equal**. Students who notice and say so deserve extra credit.

Task 2.3b • Quick percents

10% of \$60 = **\$6.00**

25% of \$80 = **\$20.00**

50% of \$37 = **\$18.50**

20% of \$45 = **\$9.00**

5% of \$120 = **\$6.00**

75% of \$32 = **\$24.00**

Task 2.3 • The jacket

Northgate: $\frac{1}{3}$ of 60 = 20 off → **\$40.00**

Riverside: 30% of 60 = 18 off → **\$42.00**

Eastway: 0.35 of 60 = 21 off → **\$39.00**

Eastway is cheapest. The point of the question: “% off” *sounds* like the biggest saving because a third feels like a lot, but 0.35 is 35%, which beats both. Students must convert to compare — there is no way to eyeball it.

Task 2.4 • Test scores

Maya $17 \div 20 = 0.85 = 85\%$ (1st)

Jordan $21 \div 25 = 0.84 = 84\%$ (2nd)

Priya $33 \div 40 = 0.825 = 82.5\%$ (3rd)

Sam $13 \div 16 = 0.8125 = 81.25\%$ (4th)

Maya is first. Note the ranks run 1–4; if a student writes 1st–4th that is correct.

Task 2.5 • both parts

a. 1/3 is larger. $1/3 = 0.3333...$ forever, which is greater than 0.333. The difference is tiny but real.

b. No — it ends at \$79.20. Up 10%: $\$80 + \$8 = \$88$. Down 10%: 10% of **\$88** is \$8.80, so $\$88 - \$8.80 = \$79.20$. The percentages are taken from different starting amounts. This is the most valuable idea on the page.

Best Buy

2 MINUTES

Students compute unit rates to find which package is actually cheaper, scale a recipe up and down, and discover that the biggest box is not always the best value. Grocery math that students immediately recognize.

You do not need to know this subject

Every answer is a division and every one is in the key, rounded and shown. Allow calculators freely — the skill is knowing what to divide by what, not the arithmetic.

Photocopy these pages only

Pages 17–19

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are recommended for this lesson.

WHAT TO SAY TO START

“The big box is always cheaper per ounce, right? Today you are going to check that, and you are going to find at least one case where it is not true. This is the math you will use in a store for the rest of your life.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read unit rate and the worked example	Do the cereal comparison on the board
14–28	Task 3.1 — unit prices	Calculators fine. Circulate
28–42	Task 3.2 — the grocery aisle	One row breaks the ‘bigger is cheaper’ rule
42–50	Task 3.3 — scaling a recipe	Multiply every ingredient
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“What do I divide by what?”

Price divided by amount gives price per unit. Always put the money on top when you want cost per ounce.

“Do I round?”

To the nearest cent, two decimal places. Say this before they start or you will get answers like \$0.0733.

“Is the bigger box always cheaper per ounce?”

No, and Task 3.2 contains a deliberate counterexample. That row is the whole point of the lesson.

If they finish early

Task 3.4 asks students to work out whether a bulk membership is worth the fee. Beyond that, ask them to find a case where buying two small packages beats one large one, and explain why a store might price it that way.

Curriculum focus • for the regular teacher’s records

Grade 6–7: ratios, unit rates and proportional reasoning; using rates to solve real-world problems. Discussion question: “Why would a store deliberately make the big box worse value than the small one?”

Cost per one

16 MIN

NAME _____

CLASS _____

DATE _____

A unit rate answers: how much for exactly one?

A **ratio** compares two amounts: 3 cups of flour to 2 cups of milk.

A **unit rate** is a ratio where the second number is 1. Miles per hour. Dollars per ounce. Cost for exactly one. It is the only fair way to compare two different-sized packages.

$$\text{unit price} = \text{price} \div \text{amount}$$

Money always goes **on top**
when you want cost per ounce.

Worked example • which cereal is better value?

Small box • 12 oz for \$3.60
 $3.60 \div 12 = \$0.30 \text{ per oz}$

Large box • 18 oz for \$4.68
 $4.68 \div 18 = \$0.26 \text{ per oz}$

The large box wins by 4 cents an ounce. Notice you cannot tell that from the prices alone — \$4.68 *looks* more expensive.

Task 3.1 • Find the unit price

DO

Round to the nearest cent.

#	ITEM	WORKING	PRICE PER UNIT
a	24 oz of pasta for \$3.12		
b	6 cans of soda for \$4.50		
c	5 lb of rice for \$8.45		
d	16 oz of juice for \$2.24		
e	3 notebooks for \$5.97		

The grocery aisle

20 MIN

Task 3.2 • Which size wins?

COMPARE

For each product, work out both unit prices and circle the better buy. Round to the nearest cent.

PRODUCT	SMALL SIZE	\$/UNIT	LARGE SIZE	\$/UNIT	BETTER BUY
Peanut butter	16 oz • \$3.20		28 oz • \$5.04		
Laundry soap	50 oz • \$8.50		100 oz • \$16.00		
Yogurt	6 oz • \$0.90		32 oz • \$5.12		
Paper towels	6 rolls • \$7.14		12 rolls • \$13.68		
Olive oil	8 oz • \$4.00		25 oz • \$13.00		

One row breaks the rule. Which product is *cheaper* in the small size? _____

Task 3.3 • Scale the recipe

PROPORTION

This recipe serves 4. Fill in the amounts for 6 servings and for 10.

INGREDIENT	SERVES 4	SERVES 6	SERVES 10
Flour (cups)	2		
Sugar (cups)	$\frac{1}{2}$		
Butter (oz)	6		
Eggs	2		
Milk (cups)	$1\frac{1}{4}$		

Which ingredient causes a problem when you scale to 6 servings, and what would you do about it in a real kitchen?

Is the membership worth it?

16 MIN

Task 3.4 • The warehouse club

DECIDE

A warehouse store charges **\$55 per year** to be a member. Members pay less per item.

ITEM	REGULAR STORE	WAREHOUSE	BOUGHT PER YEAR	YEARLY SAVING
Coffee	\$0.55/oz	\$0.40/oz	240 oz	
Detergent	\$0.17/oz	\$0.13/oz	400 oz	
Paper towels	\$1.19/roll	\$0.95/roll	60 rolls	

- a. Total saved in a year: _____
- b. After paying the \$55 fee, is the family better off? By how much? _____
- c. How much detergent would they need to buy per year for the membership to break even on detergent **alone**? _____

Task 3.5 • Think it through

CHALLENGE

Why would a store deliberately price the large size at a worse value than the small one? Give one reason a store might do this on purpose.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Best Buy

ANSWERS

Task 3.1 • Unit prices

- a. $3.12 \div 24 = \$0.13 / \text{oz}$
- b. $4.50 \div 6 = \$0.75 / \text{can}$
- c. $8.45 \div 5 = \$1.69 / \text{lb}$
- d. $2.24 \div 16 = \$0.14 / \text{oz}$
- e. $5.97 \div 3 = \$1.99 \text{ each}$

Task 3.3 • Scaling the recipe

INGREDIENT	4	6	10
Flour	2	3	5
Sugar	$\frac{1}{2}$	$\frac{3}{4}$	$1\frac{1}{4}$
Butter	6	9	15
Eggs	2	3	5
Milk	$1\frac{1}{4}$	$1\frac{1}{2}$	$2\frac{1}{4}$

Milk is the awkward one. $1\frac{1}{4}$ cups is not a measurable amount in most kitchens. A real cook would round to $1\frac{1}{2}$ or 2 cups. Accept any sensible practical answer.

Task 3.2 • The grocery aisle

PRODUCT	SMALL	LARGE	BETTER BUY
Peanut butter	\$0.20/oz	\$0.18/oz	Large
Laundry soap	\$0.17/oz	\$0.16/oz	Large
Yogurt	\$0.15/oz	\$0.16/oz	SMALL
Paper towels	\$1.19/roll	\$1.14/roll	Large
Olive oil	\$0.50/oz	\$0.52/oz	SMALL

Two rows break the rule — yogurt and olive oil. The question asks for one, so either answer is correct, but a student who finds both has done the work properly and should be told so.

Task 3.4 • The warehouse club

Coffee $(0.55 - 0.40) \times 240 = 0.15 \times 240 = \36.00

Detergent $(0.17 - 0.13) \times 400 = 0.04 \times 400 = \16.00

Towels $(1.19 - 0.95) \times 60 = 0.24 \times 60 = \14.40

a. **Total saved = \$66.40**

b. $\$66.40 - \$55.00 = \$11.40$ **better off.** Worth it, but only just — a good point to raise with the class.

c. Saving is \$0.04 per ounce, so $\$55 \div \$0.04 = 1,375$ **ounces** of detergent a year. That is roughly 11 gallons — far more than a family uses, which shows the membership only pays off across many products at once.

Task 3.5 • what to look for

Any reasonable answer: shoppers assume bigger is cheaper and stop checking · the large size may be a premium or "family" brand · the small size may be on temporary sale · the store makes more profit on the size people buy without thinking. The key idea is that **the store is relying on you not doing the division.**

Above and Below Zero

2 MINUTES

Students place integers on a number line, then add and subtract them using elevation, temperature and bank balances — three contexts where negative numbers are obviously real rather than abstract.

You do not need to know this subject

Every answer is one integer and the key shows the number-line reasoning for the tricky ones. Nothing here requires you to remember the sign rules yourself.

Photocopy these pages only

Pages 22–24

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are not needed and are best avoided here.

WHAT TO SAY TO START

“Negative numbers feel made up until you stand in Death Valley, which is 282 feet below sea level, or check a bank account that is overdrawn. Today you will work with numbers below zero the way people actually use them.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the number line panel	Do two examples on the board
12–24	Task 4.1 and 4.2 — ordering and adding	Answers in the key
24–38	Task 4.3 — the bank account	Running-total column matters
38–46	Task 4.4 — temperature changes	Subtracting a negative appears
46–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Why does subtracting a negative make it bigger?”

Taking away a debt leaves you better off. If someone cancels \$20 you owed, you gained \$20. The bank task on page 3 makes this concrete.

“Is -5 bigger or smaller than -2 ?”

Smaller. Further left on the number line is always smaller. Point at the line — it settles the argument instantly.

“Do I need a calculator?”

No. All the numbers are small on purpose, and calculators actually make sign errors harder to spot.

If they finish early

Task 4.5 introduces multiplying and dividing negatives with the sign rules. Beyond that, ask students why two negatives multiplied give a positive, using the pattern $3 \times (-2)$, $2 \times (-2)$, $1 \times (-2)$, $0 \times (-2)$... and continuing.

Curriculum focus • for the regular teacher’s records

Grade 6–7: understanding positive and negative numbers; adding and subtracting rational numbers; absolute value. Discussion question: “A temperature of -10°F and a bank balance of $-\$10$ are both ‘negative ten’. Do they mean the same kind of thing?”

Left is always less

16 MIN

NAME _____

CLASS _____

DATE _____

The number line settles every argument



Further **right** is always larger. That means -2 is larger than -8 , even though 8 is larger than 2 . When you are unsure, point at the line.

Elevation

Sea level is 0 . Death Valley is -282 ft.
Denver is $+5,280$ ft.

Temperature

Zero is just a mark on the scale. -15°F is genuinely colder than -5°F .

Money

A balance of $-\$40$ means you owe $\$40$.
Paying it off moves you toward zero.

Task 4.1 • Order them

DO

Write each set from smallest to largest.

SET	THE NUMBERS	SMALLEST $\rightarrow$ LARGEST
a	$3, -7, 0, -2, 5$	
b	$-15, -1, -20, -8$	
c	$-4, 4, -0.5, 2, -6$	

Task 4.2 • Add and subtract

DO

#	PROBLEM	ANSWER	#	PROBLEM	ANSWER
a	$-5 + 8$		e	$10 + (-14)$	
b	$3 - 9$		f	$-2 - (-8)$	
c	$-4 + (-6)$		g	$0 - 6$	
d	$-7 - (-3)$		h	$-12 + 12$	

Money makes it obvious

20 MIN

Task 4.3 • The bank account

APPLY

Ana starts the month with \$50. Fill in the running balance after each transaction.

DAY	WHAT HAPPENED	CHANGE	BALANCE
1	Starting balance	—	\$50
3	Bought sneakers	-\$68	
5	Overdraft fee	-\$15	
9	Paycheck from work	+\$40	
14	Bank cancels the fee	+\$15	
20	Bought a game	-\$25	
28	Birthday money	+\$60	

- What was her lowest balance? _____
- On which day did she get back above zero? _____
- On day 14 the bank **cancels** the \$15 fee. Explain why cancelling a charge makes the balance go **up**.

Task 4.4 • Temperature swings

APPLY

#	THE SITUATION	WORKING	ANSWER
a	It is 8°F and drops 15 degrees overnight.		
b	It is -12°F and rises 20 degrees.		
c	Monday was -3°F. Tuesday was 9°F. How much warmer?		
d	Anchorage is -6°F. Fairbanks is 22 degrees colder. What is Fairbanks?		

Multiplying with signs

16 MIN

Two rules cover everything

Same signs → positive

$$(-3) \times (-4) = +12$$

$$6 \times 2 = +12$$

Different signs → negative

$$(-3) \times 4 = -12$$

$$3 \times (-4) = -12$$

The same two rules work for division.

Task 4.5 • Multiply and divide

DO

#	PROBLEM	ANSWER	#	PROBLEM	ANSWER
a	-6×4		d	$-36 \div (-9)$	
b	$-5 \times (-7)$		e	$(-2)^3$	
c	$20 \div (-5)$		f	$(-2)^4$	

Look at e and f. Why does one come out negative and the other positive? _____

Task 4.6 • Why does it work?

CHALLENGE

Continue this pattern downward and use it to explain why $(-2) \times (-3)$ must be positive.

$$3 \times (-3) = -9$$

$$2 \times (-3) = -6$$

$$1 \times (-3) = -3$$

$$0 \times (-3) = 0$$

$$-1 \times (-3) = ______$$

$$-2 \times (-3) = ______$$

What is the pattern doing each time you go down a row? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Above and Below Zero

ANSWERS

Task 4.1 • Ordering

- a. -7, -2, 0, 3, 5
- b. -20, -15, -8, -1
- c. -6, -4, -0.5, 2, 4

Task 4.2 • Add / subtract

- a. 3
- b. -6
- c. -10
- d. -4
- e. -4
- f. 6
- g. -6
- h. 0

Task 4.5 • Signs

- a. -24
- b. 35
- c. -4
- d. 4
- e. -8
- f. 16

e vs f: an **odd** number of negatives stays negative; an **even** number cancels to positive.

Task 4.3 • Ana's account

- Day 1 \$50
- Day 3 -\$68 → -\$18
- Day 5 -\$15 → -\$33 ← lowest
- Day 9 +\$40 → \$7 ← back above zero
- Day 14 +\$15 → \$22
- Day 20 -\$25 → -\$3
- Day 28 +\$60 → \$57
- a. -\$33 b. Day 9.

c. The fee was money taken away. Cancelling it means taking away a subtraction, which puts the money back. This is exactly what $-(-15) = +15$ means, and it is the clearest explanation of subtracting a negative that students will meet.

Task 4.4 • Temperature

- a. $8 - 15 = -7^{\circ}\text{F}$
 - b. $-12 + 20 = 8^{\circ}\text{F}$
 - c. $9 - (-3) = 12$ degrees warmer
 - d. $-6 - 22 = -28^{\circ}\text{F}$
- Part c is the one to check carefully — many students answer 6.

Task 4.6 • the pattern

The missing rows are **+3** and **+6**.

Going down a row, the left factor drops by 1 and the answer **increases by 3** every single time. To keep that pattern unbroken, $-1 \times (-3)$ must be +3 and $-2 \times (-3)$ must be +6. Accept any answer that says the pattern forces it.

Keep It Balanced

2 MINUTES

Students solve one- and two-step equations using the balance-scale idea, check every answer by substituting it back, and then find the mistake in four worked solutions.

You do not need to know this subject

Every equation has one answer, and because students substitute to check, most will catch their own errors before you do. The key shows every step.

Photocopy these pages only

Pages 27–29

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are not needed.

WHAT TO SAY TO START

“An equation is a set of scales that balance. Whatever you do to one side, you must do to the other, or it tips. That single idea solves every equation you will meet this year.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the balance panel and worked example	Do one on the board, checking the answer
14–26	Task 5.1 — one-step equations	Insist on the check column
26–40	Task 5.2 — two-step equations	Undo add/subtract first
40–50	Task 5.3 — find the error	Four wrong solutions
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which do I undo first?”

Undo addition and subtraction before multiplication and division — the reverse of the order of operations. The panel on page 2 says this.

“How do I know if I’m right?”

Substitute the answer back into the original equation. If both sides come out equal, it is right. That is what the check column is for.

“What if x ends up negative?”

That is perfectly fine and happens in two of these. Nothing has gone wrong.

If they finish early

Task 5.4 asks students to write their own equation from a word problem, which is the genuinely hard skill. Beyond that, ask them to write a word problem whose answer is $x = 7$, then swap and solve.

Curriculum focus • for the regular teacher’s records

Grade 6–8: solving one-variable equations; using inverse operations; writing equations from real situations. Discussion question: “Why is checking your answer not a waste of time?”

The scales must balance

16 MIN

NAME _____

CLASS _____

DATE _____

Whatever you do to one side, do to the other



These scales balance, so $x + 3 = 8$.

To find x , take 3 off the left side. But then the scales tip — so you must take 3 off the right as well.

$$\begin{aligned} x + 3 &= 8 \\ x + 3 - 3 &= 8 - 3 \\ x &= 5 \end{aligned}$$

Check: $5 + 3 = 8$ ✓

Task 5.1 • One-step equations

DO

Solve each one, then **check** by putting your answer back in.

EQUATION	X =	CHECK
$x + 7 = 12$		
$x - 4 = 9$		
$5x = 45$		
$x \div 3 = 6$		

EQUATION	X =	CHECK
$x + 11 = 4$		
$x - 6 = -2$		
$8x = -24$		
$x/4 = 2.5$		

Task 5.1b • Which move undoes it?

THINK

For each equation, write the **single move** you would make to both sides. You do not have to solve it.

EQUATION	THE MOVE
$x + 9 = 20$	
$x - 5 = 11$	
$7x = 42$	

EQUATION	THE MOVE
$x/6 = 3$	
$x + 14 = 6$	
$-4x = 20$	

What do you notice about the move compared with what the equation does to x ? _____

Two steps, in the right order

22 MIN

Undo in reverse

To **build** $3x + 4$ you multiply first, then add. To **undo** it you go backwards: subtract first, then divide.

Undo + and – before $\times$ and $\div$.

$$3x + 4 = 19$$

$$3x + 4 - 4 = 19 - 4$$

$$3x = 15$$

$$3x \div 3 = 15 \div 3$$

$$x = 5$$

$$\text{Check: } 3(5) + 4 = 19 \checkmark$$

Task 5.2 • Two-step equations

DO

EQUATION	WORKING	X =	EQUATION	WORKING	X =
$2x + 5 = 17$			$x/2 + 7 = 11$		
$4x - 3 = 21$			$5x - 8 = -23$		
$6x + 10 = 4$			$3(x + 2) = 18$		

Task 5.3 • Find the error

DETECTIVE

Kai • $2x + 6 = 14$

$$2x = 20$$

$$x = 10$$

Error:

Correct x = _____

Lin • $5x = 30$

$$x = 30 + 5$$

$$x = 35$$

Error:

Correct x = _____

Mo • $3x - 4 = 11$

$$3x = 7$$

$$x = 7/3$$

Error:

Correct x = _____

Nia • $x/3 = 9$

$$x = 9 \div 3$$

$$x = 3$$

Error:

Correct x = _____

From words to equations

16 MIN

Task 5.4 • Write the equation, then solve it

APPLY

Let x be the unknown. Write an equation first, then solve.

#	THE SITUATION	EQUATION	$x =$
a	A number plus 12 equals 30.		
b	Four times a number is 52.		
c	Rosa has \$8 more than Ben. Together they have \$46. How much does Ben have?		
d	A taxi charges \$3 to start plus \$2 per mile. The ride cost \$23. How many miles?		
e	Three identical boxes plus a 5 lb weight balance 26 lb. What does one box weigh?		

Task 5.5 • Build one yourself

CHALLENGE

Write a word problem whose answer is exactly $x = 7$. It must need **two** steps to solve.

Swap with a partner. Did their problem really give 7? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. Why is checking your answer not a waste of time?

Answer key

Keep It Balanced

ANSWERS

Task 5.1 • One-step

$$\begin{aligned}x + 7 &= 12 \rightarrow x = 5 \\x - 4 &= 9 \rightarrow x = 13 \\5x &= 45 \rightarrow x = 9 \\x \div 3 &= 6 \rightarrow x = 18 \\x + 11 &= 4 \rightarrow x = -7 \\x - 6 &= -2 \rightarrow x = 4 \\8x &= -24 \rightarrow x = -3 \\x/4 &= 2.5 \rightarrow x = 10\end{aligned}$$

Task 5.2 • Two-step

$$\begin{aligned}2x + 5 &= 17 \rightarrow 2x = 12 \rightarrow x = 6 \\4x - 3 &= 21 \rightarrow 4x = 24 \rightarrow x = 6 \\6x + 10 &= 4 \rightarrow 6x = -6 \rightarrow x = -1 \\x/2 + 7 &= 11 \rightarrow x/2 = 4 \rightarrow x = 8 \\5x - 8 &= -23 \rightarrow 5x = -15 \rightarrow x = -3 \\3(x+2) &= 18 \rightarrow x + 2 = 6 \rightarrow x = 4\end{aligned}$$

Task 5.1b • Which move undoes it?

$$\begin{aligned}x + 9 &= 20 \rightarrow \text{subtract } 9 \\x - 5 &= 11 \rightarrow \text{add } 5 \\7x &= 42 \rightarrow \text{divide by } 7\end{aligned}$$

$$\begin{aligned}x/6 &= 3 \rightarrow \text{multiply by } 6 \\x + 14 &= 6 \rightarrow \text{subtract } 14 \\-4x &= 20 \rightarrow \text{divide by } -4\end{aligned}$$

The move is always the **opposite** of what the equation does to x . Getting a student to say that sentence out loud is worth more than extra practice.

Task 5.3 • The four errors

STUDENT	THE ERROR	CORRECT
Kai	Added 6 to both sides instead of subtracting it. Should be $2x = 14 - 6 = 8$.	$x = 4$
Lin	Added instead of dividing. $5x$ means $5 \times x$, so undo it by dividing by 5.	$x = 6$
Mo	Subtracted 4 instead of adding it. Undoing -4 means $+4$, so $3x = 15$.	$x = 5$
Nia	Divided instead of multiplying. $x/3 = 9$ means $x = 9 \times 3$.	$x = 27$

All four are the same underlying mistake: **doing the operation instead of undoing it**. Worth saying out loud once the class has found them.

Task 5.4 • Word problems

#	EQUATION	ANSWER
a	$x + 12 = 30$	$x = 18$
b	$4x = 52$	$x = 13$
c	$x + (x + 8) = 46 \rightarrow 2x + 8 = 46$	Ben has \$19
d	$2x + 3 = 23$	10 miles
e	$3x + 5 = 26$	7 lb

Part c is the hard one. Ask who the 8 belongs to — Rosa has $x + 8$, and the two **together** make 46. Rosa gets \$27, a useful check.

The last word

Substituting takes ten seconds and catches almost every slip. Accept anything reaching “it tells you whether you are right without asking anyone”.

Design a Room

2 MINUTES

Students find area and perimeter of basic shapes, split an L-shaped floor plan into rectangles, then cost out flooring and baseboard for it. The math becomes a quote a contractor would actually give.

You do not need to know this subject

Area is length times width and perimeter is the distance around. Both formulas are printed on the student page, and the key shows every calculation.

Photocopy these pages only

Pages 32–34

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil and a ruler if you have one. Calculators are fine.

WHAT TO SAY TO START

“A contractor measuring your living room has to work out two different numbers: how much carpet to buy, and how much trim to run around the edge. Today you are going to do both, and then price the job.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the formulas and worked example	Stress that area and perimeter answer different questions
14–26	Task 6.1 — basic shapes	Watch the units
26–42	Task 6.2 — the L-shaped plan	Split into two rectangles
42–52	Task 6.3 — cost the job	Multiply by price per unit
52–60	Challenge, collect sheets	Discussion question below

Questions students will ask

“Do I add or multiply?”

Multiply for area, add for perimeter. Area covers the floor, perimeter runs around the edge.

“How do I do the L shape?”

Draw one line to cut it into two rectangles, find each area, then add. The dashed line on the diagram shows one way.

“Why is it square feet and not feet?”

Because you multiplied two lengths together. Perimeter is just feet because you only added lengths.

If they finish early

Task 6.4 asks students to design a room to a fixed budget, which has many valid answers. Beyond that, ask them to find two different rectangles with the same perimeter but different areas — it surprises most students that this is possible.

Curriculum focus • for the regular teacher’s records

Grade 6–7: area of triangles, quadrilaterals and composite figures; real-world problems involving area and perimeter. Discussion question: “Two rooms have the same perimeter. Must they have the same area?”

Two different questions

16 MIN

NAME _____

CLASS _____

DATE _____

Area covers it. Perimeter goes around it.

AREA = length $\times$ width
answer in **square** units (ft²)
→ carpet, paint, turf

PERIMETER = add every side
answer in **plain** units (ft)
→ fencing, trim, baseboard

Triangle area = $\frac{1}{2} \times \text{base} \times \text{height}$. A triangle is half of the rectangle that would fit around it.

Task 6.1 • Find both

DO

Give the correct units for each answer.

#	SHAPE	WORKING	AREA	PERIMETER
a	Rectangle 8 ft $\times$ 5 ft			
b	Square, side 9 ft			
c	Rectangle 12 ft $\times$ 4 ft			
d	Rectangle 7.5 ft $\times$ 6 ft			
e	Triangle, base 10 ft, height 6 ft			—

Watch this

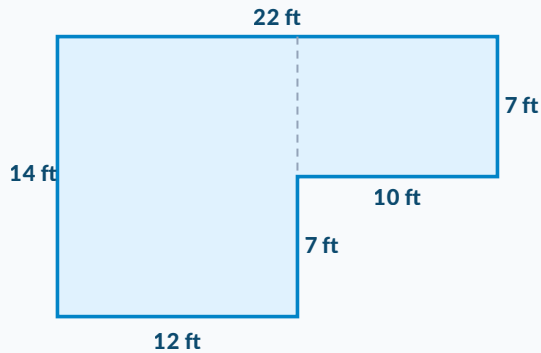
A rectangle 8 $\times$ 5 has area 40 and perimeter 26. Those are different numbers answering different questions. Getting them mixed up is the single most common mistake in this topic.

A real floor plan

22 MIN

Task 6.2 • The L-shaped living room

APPLY



The dashed line shows one way to split it into two rectangles. You may split it differently if you prefer.

a. Rectangle 1: _____ ft $\times$ _____ ft = _____ ft^2

b. Rectangle 2: _____ ft $\times$ _____ ft = _____ ft^2

c. Total floor area = _____ ft^2

d. Add up all six outside walls to find the perimeter.

Perimeter = _____ ft

Task 6.3 • Price the job

COST

Use your answers from Task 6.2.

MATERIAL	PRICE	HOW MUCH IS NEEDED	WORKING	COST
Carpet	\$3.50 per ft^2	the floor area		
Baseboard	\$2.25 per ft	the perimeter		
Underlay	\$0.80 per ft^2	the floor area		

Total cost of the job: _____

Carpet is sold only in whole square feet and you should order 10% extra for waste. How much would you actually order? _____

Task 6.4 • Your own room

CREATE

Design a rectangular bedroom with a floor area between **120 and 150 ft²**. Sketch it, label the sides, then cost the carpet at \$3.50 per ft².



Length _____ ft Width _____ ft Area _____ ft² Perimeter _____ ft

Carpet cost: _____

Task 6.5 • Same edge, different floor

CHALLENGE

Both rooms below have a perimeter of **24 ft**. Find the area of each.

ROOM	DIMENSIONS	PERIMETER	AREA
A	10 ft × 2 ft	24 ft	
B	8 ft × 4 ft	24 ft	
C	6 ft × 6 ft	24 ft	

Which shape gives the **most** floor for the same amount of wall? _____

Why might a builder care about that?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Design a Room

ANSWERS

Task 6.1 • Area and perimeter

#	SHAPE	AREA	PERIMETER
a	8×5	40 ft^2	26 ft
b	9×9	81 ft^2	36 ft
c	12×4	48 ft^2	32 ft
d	7.5×6	45 ft^2	27 ft
e	$\frac{1}{2} \times 10 \times 6$	30 ft^2	—

Task 6.2 • The L-shaped room

Split along the dashed line

Rectangle 1 (top): $22 \times 7 = 154$

Rectangle 2 (left): $12 \times 7 = 84$

Total area = 238 ft^2

Perimeter • all six sides

$22 + 7 + 10 + 7 + 12 + 14$

= 72 ft

Other splits give the same total — accept any correct decomposition. If a student's two rectangles do not sum to 238, check their side lengths against the diagram before marking it wrong.

Task 6.3 • The quote

Carpet $238 \times 3.50 = \$833.00$

Baseboard $72 \times 2.25 = \$162.00$

Underlay $238 \times 0.80 = \$190.40$

Total = $\$1,185.40$

With 10% waste: $238 \times 1.1 = 261.8$, so order 262 ft^2 of carpet.

Task 6.5 • Same perimeter, different area

A $10 \times 2 = 20 \text{ ft}^2$

B $8 \times 4 = 32 \text{ ft}^2$

C $6 \times 6 = 36 \text{ ft}^2$

The **square** gives the most floor for the same wall. A builder cares because walls cost money and floor space is what people pay for — a square room is the cheapest way to buy area.

Grading Task 6.4

Any rectangle with area between 120 and 150 ft^2 is correct — 10×13 , 12×12 , 11×12 and many others. Check that their stated area actually equals length $\times$ width, and that the carpet cost equals area $\times 3.50$.

How Graphs Lie

2 MINUTES

Students compute mean, median, mode and range, work out which average is fairest for a set of salaries, then compare two bar charts of identical data that tell completely different stories.

You do not need to know this subject

The calculations are addition and division, all in the key. The graph task is a judgment question and the key explains exactly what to look for.

Photocopy these pages only

Pages 37–39

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are fine for the mean.

WHAT TO SAY TO START

“Two graphs on this sheet show the same three numbers. One makes them look almost identical and the other makes one look twice the size of another. Neither is lying about the data. Today you find out how that is possible.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the four measures and the example	Order the data first — always
14–26	Task 7.1 — compute all four	Straight practice
26–38	Task 7.2 — the salary problem	Which average is honest? Great discussion
38–50	Task 7.3 — the two graphs	The payoff of the lesson
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“What if there are two middle numbers?”

Add them and divide by two. The panel on page 1 says this, but expect to be asked.

“What if no number repeats?”

Then there is no mode. That is a legitimate answer, not a mistake.

“Which average is the right one?”

It depends what you are trying to show — which is the entire point of Task 7.2. There is no single right answer and the key explains why.

If they finish early

Task 7.4 asks students to redraw the misleading graph honestly. Beyond that, ask them to find a graph in a newspaper or online and check whether its vertical axis starts at zero.

Curriculum focus • for the regular teacher’s records

Grade 6–7: measures of center and variability; summarizing numerical data; displaying data and evaluating displays. Discussion question: *“If a graph is technically accurate but leaves the wrong impression, is it a lie?”*

Four ways to describe a set

16 MIN

NAME _____

CLASS _____

DATE _____

Always put the numbers in order first

MEAN

Add them all, divide by how many.

MEDIAN

The middle one, once ordered.
Two in the middle? Average them.

MODE

The one that appears most.
There may be none.

RANGE

Largest minus smallest. How spread out it is.

Worked example • 7, 3, 9, 3, 8

Ordered: 3, 3, 7, 8, 9

Mean = $(3+3+7+8+9) \div 5 = 30 \div 5 = 6$

Median = middle of five = 7

Mode = 3 (appears twice)

Range = $9 - 3 = 6$

Task 7.1 • All four, for each set

DO

SET	THE DATA	MEAN	MEDIAN	MODE	RANGE
A	4, 8, 6, 8, 4, 8				
B	12, 15, 11, 18				
C	5, 5, 20, 5, 5				

Look at set C. Which measure gives the most **misleading** picture of that data, and why? _____

Which average is honest?

18 MIN

Task 7.2 • Salaries at a small company

JUDGE

Nine people work at Vertex Design. These are their yearly salaries.

\$34,000 \$36,000 \$38,000 \$38,000 \$40,000
\$42,000 \$45,000 \$48,000 \$420,000 (the owner)

a. Mean salary: _____

b. Median salary: _____

c. Mode: _____

d. The company's hiring ad says "average salary over \$78,000". Which average did they use, and is the ad lying?

e. Which average would you use if you wanted to tell a job applicant the truth, and why? _____

Task 7.2b • Take the owner out

INVESTIGATE

Now remove the \$420,000 salary and use only the other eight.

	WITH THE OWNER	WITHOUT THE OWNER	HOW MUCH DID IT CHANGE?
Mean			
Median			

a. Which measure barely moved? _____

b. A value far away from all the others is called an **outlier**. Which measure is more affected by an outlier, and why does that matter when somebody quotes an average at you?

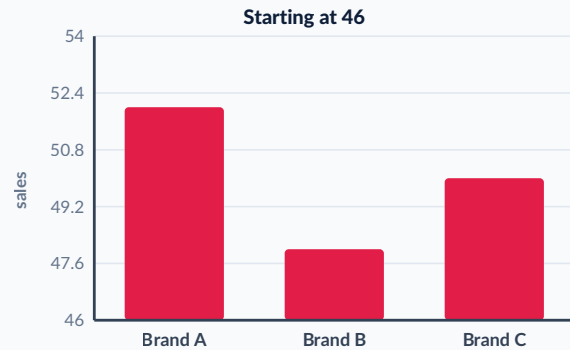
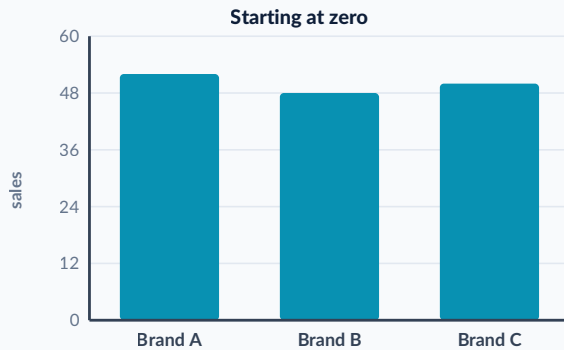
The same data, twice

18 MIN

Task 7.3 • Two graphs, identical numbers

INVESTIGATE

Brand A sold 52. Brand B sold 48. Brand C sold 50. Both graphs below show exactly those numbers.



- In the right-hand graph, Brand A's bar looks about how many times taller than Brand B's? _____
- In reality, how much bigger is A than B? _____
- What exactly did the second graph change? _____
- If you worked in marketing for Brand A, which graph would you put in the ad, and why?

Task 7.4 • The one question to always ask

CHALLENGE

Write the single question you should ask yourself every time you see a bar graph.

Name one situation where **not** starting at zero is actually reasonable. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. If a graph is technically accurate but leaves the wrong impression, is it a lie?

Answer key

How Graphs Lie

ANSWERS

Task 7.1 • The three sets

SET	ORDERED	MEAN	MEDIAN	MODE	RANGE
A	4, 4, 6, 8, 8, 8	6.33	7	8	4
B	11, 12, 15, 18	14	13.5	none	7
C	5, 5, 5, 5, 20	8	5	5	15

Set C: the **mean of 8** is the misleading one. Four of the five values are 5, so 8 describes nobody. The single value of 20 has dragged it up. The median of 5 is far more honest – and this is exactly the idea Task 7.2 builds on.

Task 7.2 • Vertex Design

Total = \$741,000

a. Mean = \$82,333

b. Median = \$40,000

c. Mode = \$38,000

d. They used the **mean**. The ad is **technically true and deeply misleading**. Eight of the nine employees earn less than half of it. One salary of \$420,000 pulls the mean up on its own.

e. The **median, \$40,000**. It describes what a typical employee actually earns and is not affected by one extreme value.

Task 7.2b • Removing the outlier

	WITH OWNER	WITHOUT	CHANGE
Mean	\$82,333	\$40,125	–\$42,208
Median	\$40,000	\$39,000	–\$1,000

a. The **median** barely moved – it fell by \$1,000 while the mean collapsed by over \$42,000. b. The **mean** is far more affected, because every value is added in, so one enormous number drags the whole thing up. When somebody quotes an “average” at you, the useful question is *which one, and is there an outlier?*

Task 7.3 • The two graphs

- Roughly **three times** taller – accept anything from 2½ to 4.
- Only **4 more sales**, about 8% bigger.
- It **started the vertical axis at 46 instead of 0**. No number was altered. Chopping off the bottom of every bar exaggerates the differences that remain.
- The right-hand one, obviously – it makes a tiny lead look commanding. Accept any answer that names the advantage honestly.

Task 7.4 • the question

“Where does the vertical axis start?” Accept any wording that checks the baseline. It is the single most useful habit in reading data, and students who take it away have gotten the point of the lesson.

When a non-zero axis is fair

Where zero is meaningless or the variation is genuinely the story: body temperature, stock prices over one day, a marathon runner’s pace. Nobody wants a temperature chart starting at absolute zero. The rule is a **prompt to check**, not an absolute law.

Is the Game Fair?

2 MINUTES

Students calculate probabilities for spinners and dice, build the two-dice sum table, then decide whether a carnival game is fair — and work out how much the operator expects to make.

You do not need to know this subject

Probabilities here are simple fractions and the key gives every one. The dice table is printed complete in the key so you can check it at a glance.

Photocopy these pages only

Pages 42–44

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Real dice are a bonus but are not required.

WHAT TO SAY TO START

“Every carnival game in the world is designed so the operator wins. Today you are going to prove that with math, and then design a game that is actually fair.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the probability panel	Do the spinner together
14–26	Task 8.1 — spinners and one die	Simple fractions
26–40	Task 8.2 — the two-dice table	36 outcomes. Insist they fill it in
40–50	Task 8.3 — the carnival game	The payoff task
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is 7 really the most likely?”

Yes, and the table proves it — six of the 36 combinations give 7, more than any other total. Students find this genuinely surprising.

“Aren't there only 11 outcomes with two dice?”

There are 11 possible totals but 36 possible outcomes, because a 2 and a 5 is a different roll from a 5 and a 2. That distinction is the whole lesson.

“Does a losing streak mean I'm due a win?”

No. Dice have no memory. This comes up every time and is worth answering clearly.

If they finish early

Task 8.4 asks students to design a fair game, which has many valid answers. Beyond that, ask them to alter the carnival game's payout so the operator breaks exactly even, and then explain why no real carnival would ever run it.

Curriculum focus • for the regular teacher's records

Grade 7: probability of simple and compound events; developing probability models; using organized lists and tables to find sample space. Discussion question: “If a game is fair, why would anybody run it?”

Counting what could happen

16 MIN

NAME _____

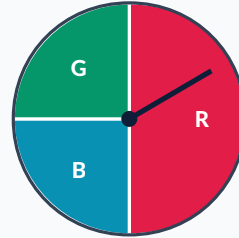
CLASS _____

DATE _____

Probability is just a fraction

$$P(\text{event}) = \frac{\text{ways it can happen}}{\text{total possible outcomes}}$$

0 means impossible. 1 means certain. Everything real sits between.



On this spinner, red covers half the circle, blue a quarter, green a quarter. $P(\text{red}) = \frac{1}{2}$, $P(\text{blue}) = \frac{1}{4}$.

Task 8.1 • Basic probabilities

DO

Give each answer as a fraction in its simplest form.

ROLLING ONE DIE	PROBABILITY	THE SPINNER ABOVE	PROBABILITY
P(roll a 4)		P(red)	
P(roll an even number)		P(not red)	
P(roll more than 4)		P(blue or green)	
P(roll a 7)		P(yellow)	

Two of these answers are the same number. Which two, and does that make sense? _____

Two dice change everything

20 MIN

Task 8.2 • Build the sum table

DO

Fill in the total for every combination of two dice. There are **36** boxes because a 2-then-5 is a different roll from a 5-then-2.

+	1	2	3	4	5	6
1	2					
2						
3						
4						
5						
6						12

TOTAL	HOW MANY WAYS	PROBABILITY	TOTAL	HOW MANY WAYS	PROBABILITY
2			10		
5			12		
7			7 or 11		

Which total is the most likely, and why is it that one? _____

Task 8.2b • Two coins

LIST

Flip two coins. List **every** possible outcome, then answer the questions.

COIN 1	COIN 2

a. How many outcomes altogether? _____

b. $P(\text{two heads}) =$ _____

c. $P(\text{one head and one tail}) =$ _____

d. Many people say the answer to c is $\frac{1}{3}$. Why are they wrong?

The carnival game

16 MIN

Task 8.3 • Would you play?

DECIDE

A stall charges **\$2 to play**. You roll two dice. If you get **7 or 11** you win **\$8**. Anything else and you lose your \$2.

- a. Using your table, $P(7 \text{ or } 11) =$ _____
- b. Out of 36 plays, how many would you expect to win? _____

Now imagine 36 people each play once.

	WORKING	AMOUNT
The stall collects	$36 \times \$2$	
The stall pays out		
The stall keeps		

- c. On average, how much does the stall make per player? _____
- d. Is this game fair? Explain using your numbers.

Task 8.4 • Make it fair

CHALLENGE

Change **one** thing about the game so it becomes fair – so that over many plays, the stall and the players break even.

If the game were genuinely fair, why would nobody set up the stall? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Is the Game Fair?

ANSWERS

Task 8.1 • Basic probabilities

$$P(4) = 1/6$$

$$P(\text{even}) = 3/6 = 1/2$$

$$P(\text{more than } 4) = 2/6 = 1/3$$

$$P(7) = 0$$

$$P(\text{red}) = 1/2$$

$$P(\text{not red}) = 1/2$$

$$P(\text{blue or green}) = 1/2$$

$$P(\text{yellow}) = 0$$

P(not red) and P(blue or green) are both 1/2 — and they must be, because not-red is blue-or-green. Two descriptions of the same event.

Task 8.2 • The counts

TOTAL	WAYS	PROBABILITY
2	1	1/36
5	4	1/9
7	6	1/6
10	3	1/12
12	1	1/36
7 or 11	8	2/9

7 is most likely — six combinations give it, more than any other total.

Task 8.2b • two coins. Outcomes are HH HT TH TT — 4 of them. $P(\text{two heads}) = 1/4$. $P(\text{one of each}) = 2/4 = 1/2$. Students who say 1/3 have counted HT and TH as one outcome; the coins are distinguishable, so they are two — the same idea as 2+5 and 5+2 being different rolls.

Task 8.2 • Completed sum table

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Task 8.3 • The carnival game

a. $P(7 \text{ or } 11) = 8/36 = 2/9$

b. About **8 wins** in 36 plays

Collects $36 \times \$2 = \72

Pays out $8 \times \$8 = \64

Keeps **\$8**

c. $\$8 \div 36 =$ about **22 cents per player**.

d. **Not fair.** The stall keeps money on average, so players lose. Winning \$8 on a \$2 stake looks generous, but the win is too rare. Any answer using \$72 against \$64 earns full credit.

Task 8.4 • what makes it fair

Cleanest fix: **raise the prize to \$9** — $8 \times \$9 = \72 , matching the \$72 collected. Also accept a lower entry fee or more winning totals. And nobody would run it because **a fair game makes zero profit**. Every real carnival game has a built-in edge.

The Coordinate Plane

2 MINUTES

Students name the quadrants, read and plot ordered pairs, connect a set of points into a shape, then reflect and translate it. Very visual, and mistakes are immediately obvious on the page.

You do not need to know this subject

Plotting is a mechanical skill — across first, then up or down. The key gives every coordinate and describes the shape students should end up with.

Photocopy these pages only

Pages 47–49

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil and a ruler if you have one.

WHAT TO SAY TO START

“Every map app, every video game and every seat number in a stadium uses the same idea: two numbers locate one exact spot. Today you will get very good at reading and writing those two numbers.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the quadrants panel	Across first, then up. Say it twice
14–26	Task 9.1 — read and plot points	Errors are visible on the grid
26–40	Task 9.2 — the mystery shape	Connecting in order matters
40–50	Task 9.3 — reflect and translate	Watch the signs
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which number comes first?”

The x, and x is across. “Across the hallway, then up the stairs” is the phrase most students remember.

“Which quadrant is which?”

They are numbered counterclockwise starting from the top right. The diagram on page 1 shows it.

“Do I connect the points in order?”

Yes, in the order listed, and join the last one back to the first. Otherwise the shape will not appear.

If they finish early

Task 9.4 asks students to design their own shape and write the coordinates for a partner to plot. Beyond that, ask them what happens to a point when both signs are flipped, and which single transformation that is equivalent to.

Curriculum focus • for the regular teacher’s records

Grade 6–8: graphing points in all four quadrants; solving problems by graphing; reflections and translations in the plane. Discussion question: “Why does the order of the two numbers matter so much?”

Two numbers, one spot

16 MIN

NAME _____

CLASS _____

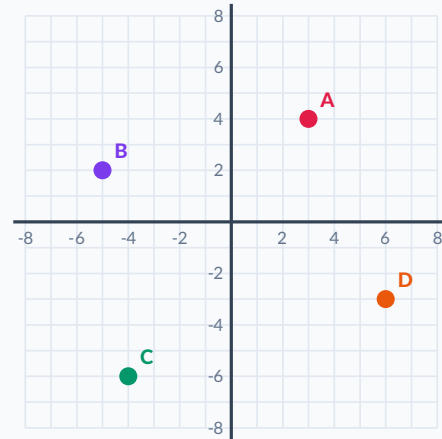
DATE _____

Across first, then up or down

A point is written (x, y) . The **x** tells you how far across; the **y** tells you how far up or down. Always in that order.

$(3, 4) \rightarrow 3$ right, 4 up
 $(-5, 2) \rightarrow 5$ left, 2 up
 $(-4, -6) \rightarrow 4$ left, 6 down
 $(6, -3) \rightarrow 6$ right, 3 down

Quadrants are numbered **counterclockwise** from the top right: I, II, III, IV.



Task 9.1 • Read and write

DO

Using the grid above, write the coordinates and the quadrant.

POINT	COORDINATES	QUADRANT
A		
B		
C		
D		

In which quadrant does each of these sit? No plotting needed.

POINT	QUADRANT
$(7, 2)$	
$(-3, -8)$	
$(-6, 5)$	
$(0, 4)$	

One of those four is a trick. Which, and why?

Plot it and see

20 MIN

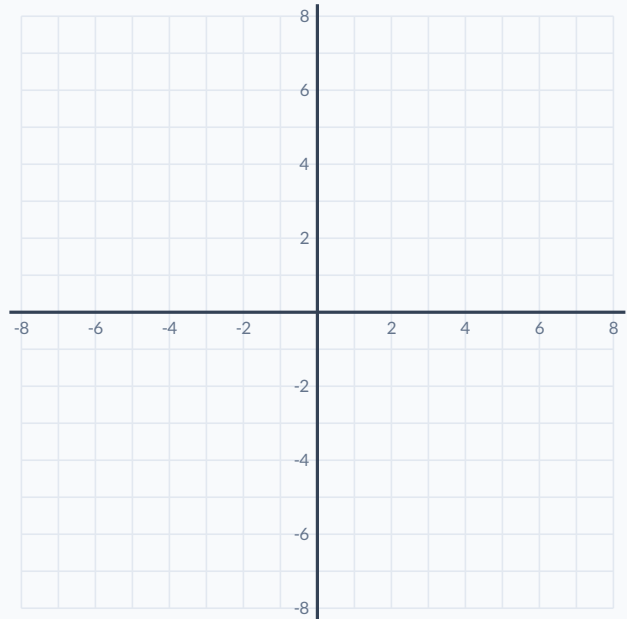
Task 9.2 • The mystery shape

PLOT

Plot these points **in order** and join each to the next. Join the last one back to the first.

$(-6, 2) \rightarrow (-2, 6) \rightarrow (2, 6)$
 $(6, 2) \rightarrow (6, -2) \rightarrow (2, -6)$
 $(-2, -6) \rightarrow (-6, -2) \rightarrow \text{back to start}$

- What shape did you draw? _____
- How many sides does it have? _____
- Is it symmetrical? How can you tell from the **coordinates** alone, without looking at your drawing?



Task 9.2b • Distance without a ruler

CALCULATE

When two points share an x-value or a y-value, you can find the distance between them by **subtracting** — no ruler needed.

THE TWO POINTS

DISTANCE

$(2, 1)$ and $(2, 7)$

$(-3, 4)$ and $(5, 4)$

$(-6, -2)$ and $(-6, 5)$

A rectangle has corners at $(-3, 2)$, $(5, 2)$, $(5, -3)$ and $(-3, -3)$.

Width _____ Height _____

Perimeter _____ Area _____

Why does subtracting work here but not for any two random points?

Two transformations

TRANSLATE • slide it
right 3, up 2 → add to both
 $(1, 4) \rightarrow (1+3, 4+2) = (4, 6)$

REFLECT • flip it
over the y-axis → flip the x sign
 $(1, 4) \rightarrow (-1, 4)$

Reflecting over the **x-axis** flips the **y** sign instead: $(1, 4) \rightarrow (1, -4)$.

Task 9.3 • Transform each point

DO

START	TRANSLATE RIGHT 4, DOWN 3	REFLECT OVER Y-AXIS	REFLECT OVER X-AXIS
$(2, 5)$			
$(-3, 1)$			
$(0, -4)$			
$(-6, -2)$			

Task 9.4 • Both at once

CHALLENGE

a. Take $(5, 3)$ and flip **both** signs to get $(-5, -3)$. Describe that single move in words.

b. Write four coordinates that would draw a rectangle for a partner to plot. Do not draw it — just list the points.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

The Coordinate Plane

ANSWERS

Task 9.1 • Reading the grid

A = (3, 4) → Quadrant I

B = (-5, 2) → Quadrant II

C = (-4, -6) → Quadrant III

D = (6, -3) → Quadrant IV

(7, 2) → I

(-3, -8) → III

(-6, 5) → II

(0, 4) → none

(0, 4) is the trick. It sits on the y-axis, and the axes belong to no quadrant.

Task 9.3 • Transformations

START	+4, -3	FLIP Y-AXIS	FLIP X-AXIS
(2, 5)	(6, 2)	(-2, 5)	(2, -5)
(-3, 1)	(1, -2)	(3, 1)	(-3, -1)
(0, -4)	(4, -7)	(0, -4)	(0, 4)
(-6, -2)	(-2, -5)	(6, -2)	(-6, 2)

(0, -4) reflected over the y-axis is **unchanged**, because it already sits on that axis.

Task 9.2b • Distance by subtraction

(2, 1) - (2, 7) → 7 - 1 = 6

(-3, 4) - (5, 4) → 5 - (-3) = 8

(-6, -2) - (-6, 5) → 5 - (-2) = 7

Rectangle: width **8**, height **5**, perimeter **26**, area **40**.

Subtracting only works when the two points share an x or a y, so the segment is **horizontal or vertical**. For a diagonal you would need the Pythagorean theorem — a good preview of what comes next.

Task 9.2 • The mystery shape

a. A regular **octagon** — accept "stop sign" or "eight-sided shape".

b. Eight sides.

c. Yes, and you can tell from the coordinates alone: **every point appears with its signs flipped somewhere else in the list**. (-6, 2) has a partner at (6, -2); (-2, 6) pairs with (2, -6). That mirroring in the numbers is the symmetry. A student who spots this without looking at their drawing has done something genuinely sophisticated.

Task 9.4a • flipping both signs

It is a **rotation of 180° about the origin** — turning the point halfway around the center. Also accept "reflecting over both axes, one after the other", which produces exactly the same result and is a good observation.

Task 9.4b • grading the rectangle

Any four points forming a rectangle. Check that two share one x-value, two share the other, and the same for y — for example (2, 1), (7, 1), (7, 5), (2, 5). A common error is listing four points that make a parallelogram or nothing at all; have the partner plot it, which settles it instantly.

Tips, Discounts & Sales Tax

2 MINUTES

Students calculate tips, discounts and sales tax, then handle a full restaurant bill and a shopping trip where the discount and the tax both apply. The most immediately useful lesson in the pack.

You do not need to know this subject

Everything is percent of a number, which is multiplication. The key shows the working for every part, including the two questions students most often get wrong.

Photocopy these pages only

Pages 52–54

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Calculators are fine and realistic for this topic.

WHAT TO SAY TO START

“Every one of you will use this math within a week of leaving this room. Tips, sale prices, sales tax. Today you are going to get fast at it, and you will find out about one trick stores use that most adults still fall for.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the percent panel	Show the 10% shortcut on the board
14–26	Task 10.1 – tips	The 10% trick makes this fast
26–38	Task 10.2 – discounts	Includes the double-discount trap
38–52	Task 10.3 and 10.4 – tax and the full bill	Order of operations matters here
52–60	Challenge, collect sheets	Discussion question below

Questions students will ask

“How do I find 15%?”

Find 10% by moving the decimal one place left, then halve it to get 5%, then add. This shortcut is on the student page and is worth demonstrating.

“Do I add tax before or after the discount?”

Discount first, then tax on the reduced price. That is how a real register does it, and Task 10.4 depends on it.

“Is 20% off then 10% off the same as 30% off?”

No, and that is the whole point of Task 10.2c. Let them discover it rather than telling them.

If they finish early

Task 10.5 asks students to work backwards from a final price to the original. Beyond that, ask them to calculate the tip on the pre-tax amount versus the post-tax amount and say which is customary and why.

Curriculum focus • for the regular teacher’s records

Grade 6–7: percent of a quantity; solving percent problems including tax, tip, markup and discount. Discussion question: “Why do stores advertise ‘20% off plus an extra 10%’ instead of just saying 28% off?”

Percent of an amount

16 MIN

NAME _____

CLASS _____

DATE _____

Two ways, and one shortcut worth knowing

The reliable way

$$\begin{aligned} &15\% \text{ of } \$40 \\ &= 0.15 \times 40 \\ &= \$6.00 \end{aligned}$$

The mental shortcut

$$\begin{aligned} &10\% \text{ of } 40 = 4.00 \text{ (move the point)} \\ &5\% \text{ is half of that} = 2.00 \\ &15\% = 4.00 + 2.00 = \$6.00 \end{aligned}$$

10% is always just the decimal point moved one place left. Everything else builds from there: 5% is half of 10%, 20% is double it.

Task 10.1 • Leave the tip

DO

Round to the nearest cent.

#	THE BILL	TIP RATE	WORKING	TIP	TOTAL PAID
a	\$40.00	15%			
b	\$26.00	20%			
c	\$85.00	18%			
d	\$62.50	20%			

Which of these could you do in your head using the 10% trick, and which needed a calculator? _____

Task 10.1b • Mental math drill

DRILL

No calculator for this one. Use 10% first, then build.

FIND	ANSWER	FIND	ANSWER
10% of \$70		10% of \$24	
5% of \$70		30% of \$24	
15% of \$70		1% of \$24	
20% of \$70		25% of \$24	

How do you get 1% from 10%? _____

Sale prices and the trap

20 MIN

Task 10.2 • Work out the sale price

DO

Find the amount off, then the price you actually pay.

#	ORIGINAL	DISCOUNT	AMOUNT OFF	YOU PAY
a	\$60.00	25% off		
b	\$48.00	30% off		
c	\$120.00	40% off		
d	\$35.00	15% off		

The trap. A jacket costs **\$100**. A sign says “20% off, plus take an extra 10% off at the register.”

- e. Price after the 20% is taken off: _____
- f. Now take 10% off **that** price. Final price: _____
- g. What would you have paid with a straight 30% off? _____
- h. Are they the same? Explain what is going on.

Task 10.3 • Add the sales tax

DO

Sales tax where you live is **7%**. Find the tax and the total.

#	PRICE	WORKING	TAX	TOTAL
a	\$20.00			
b	\$45.00			
c	\$8.50			

Put it all together

18 MIN

Task 10.4 • The whole restaurant bill

APPLY

Four friends eat out. Sales tax is **7%**. They tip **18%** on the food before tax, which is the usual custom.

ITEM	AMOUNT
Food and drinks	\$86.00
Coupon: 10% off the food	
Food after the coupon	
Sales tax at 7% of the discounted food	
Tip at 18% of the discounted food	
Total to pay	
Each person's share, split four ways	

Why is the tip worked out on the food rather than on the food plus tax? _____

Task 10.5 • Work backwards

CHALLENGE

a. A shirt is on sale for **\$36** after 25% off. What was the original price?

b. A meal came to **\$53.50** including 7% tax. What was the pre-tax price? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. Why do stores advertise “20% off plus an extra 10%” instead of just saying 28% off?

Answer key

Tips, Discounts & Sales Tax

ANSWERS

Task 10.1 • Tips

- a. \$6.00 → \$46.00
- b. \$5.20 → \$31.20
- c. \$15.30 → \$100.30
- d. \$12.50 → \$75.00

Task 10.2 • Discounts

- a. \$15 off → \$45.00
- b. \$14.40 off → \$33.60
- c. \$48 off → \$72.00
- d. \$5.25 off → \$29.75

Task 10.3 • Sales tax

- a. \$1.40 → \$21.40
- b. \$3.15 → \$48.15
- c. \$0.60 → \$9.10
- c rounds from \$0.595.

Task 10.1b • Mental math drill

$$10\% \text{ of } \$70 = \$7.00$$

$$5\% \text{ of } \$70 = \$3.50$$

$$15\% \text{ of } \$70 = \$10.50$$

$$20\% \text{ of } \$70 = \$14.00$$

1% from 10%: divide by 10 again — move the decimal point one more place left. That single trick gets students to any percent they will ever need.

$$10\% \text{ of } \$24 = \$2.40$$

$$30\% \text{ of } \$24 = \$7.20$$

$$1\% \text{ of } \$24 = \$0.24$$

$$25\% \text{ of } \$24 = \$6.00$$

Task 10.2 • The stacked-discount trap

$$\text{e. } \$100 - 20\% = \$80.00$$

$$\text{f. } \$80 - 10\% \text{ of } 80 = 80 - 8 = \$72.00$$

$$\text{g. } \$100 - 30\% = \$70.00$$

h. No — stacking is worse for the shopper. The second discount is taken from the **already reduced** \$80, not from the original \$100, so it is only worth \$8 instead of \$10. The two discounts together come to **28% off**, not 30%. This is the most valuable idea in the lesson and is worth putting on the board.

Task 10.4 • The restaurant bill

Food	\$86.00
Coupon 10%	-\$8.60
After coupon	\$77.40
Tax 7% of 77.40	\$5.42
Tip 18% of 77.40	\$13.93
Total	\$96.75
Each of four	\$24.19

Tax rounds from \$5.418, tip from \$13.932, and the split from \$24.1875. Accept answers a cent either way. **Why tip on the food?** Because the tax goes to the government, not the server — tipping on it would mean tipping on money the restaurant never receives.

Task 10.5 • working backwards

a. \$36 is 75% of the original, so original = $36 \div 0.75 = \$48.00$. **b.** \$53.50 is 107% of the pre-tax price, so pre-tax = $53.50 \div 1.07 = \$50.00$. Both require **dividing** rather than multiplying, which is why they are hard. A student who tries taking 25% off \$36 gets \$27 — point out that this makes the answer smaller when it should be bigger.

The last word

Two stacked discounts **sound** larger than one, and they are always smaller than the sum. “20% plus 10%” reads as 30 but delivers 28. Accept any answer reaching “it sounds bigger than it is”.

Handover note

90 SECONDS

Copy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE
_____CLASS
_____DATE
_____PERIOD
_____**Lesson used**

CHECK ONE

- | | |
|--|---|
| ☐ 01 • The Answer Is Wrong | ☐ 06 • Design a Room |
| ☐ 02 • Fractions, Decimals & Percents | ☐ 07 • How Graphs Lie |
| ☐ 03 • Best Buy | ☐ 08 • Is the Game Fair? |
| ☐ 04 • Above and Below Zero | ☐ 09 • The Coordinate Plane |
| ☐ 05 • Keep It Balanced | ☐ 10 • Tips, Discounts & Sales Tax |

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention • for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is what makes the next teacher's day work, and it is genuinely appreciated.

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- Solving one- and two-step equations
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- Percent applications: tip, discount and tax

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